

Stakeholder Portal — Implementation Plan

Overview

Build out the admin dashboard API to support role-based stakeholder views for Trustees, Asset Custodians, and Asset Managers. The existing codebase already handles organization management, member auth, wallet linkage, and stakeholder type linking — this plan builds on top of that.

Suggested priority: Phase 1 -> 2 -> 3 -> 5 -> 4 -> 6 -> 7 (Trustee first because they're the approval bottleneck; Asset Manager before Custodian because they initiate most workflows.)

Phase 1: Foundation — RBAC & Stakeholder Type Routing

Goal: Gate endpoints by stakeholder type so each role only accesses their own pages.

Tasks

- 1.1 Extend OrganizationAuthMiddleware to inject stakeholder_type into request context (read from org's linked stakeholder)
- 1.2 Create RequireStakeholderType(types ...string) middleware that checks context and returns 403 if role doesn't match
- 1.3 Add route groups per stakeholder type in the organizations controller:
 - /api/v1/stakeholder/trustee/...
 - /api/v1/stakeholder/custodian/...
 - /api/v1/stakeholder/asset-manager/...
 - /api/v1/stakeholder/shared/... (cross-functional)
- 1.4 Map existing stakeholder type constants to the three dashboard roles:
 - trustees -> Trustee dashboard
 - approved_asset_custodian -> Custodian dashboard
 - asset_manager -> Asset Manager dashboard

Files to modify

- internal/middleware/ — new role-checking middleware
 - internal/components/organizations/controllers/main.go — new route groups
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Phase 2: Shared Tokenized Assets Module

Goal: Unified asset listing and detail view accessible to all three roles, with role-appropriate actions.

Tasks

- 2.1 GET /api/v1/stakeholder/shared/assets — list tokenized assets
 - Paginated, filterable by asset class, status, date range, assigned stakeholder
 - Uses existing trovosdk.GetTokenizedAssetData() as data source
 - Response includes role-specific action flags (canApprove, canProcessRelease, canUpdateStatus)
- 2.2 GET /api/v1/stakeholder/shared/assets/:asset_id — asset detail
 - Full asset info: description, financials, operational status, token holders, documents, activity history
 - Role-specific action buttons in response metadata
- 2.3 Create stakeholder_assets service layer to fetch and enrich asset data per role

New files

- internal/components/stakeholder/ — new component directory
 - internal/components/stakeholder/handlers/asset_handler.go
 - internal/components/stakeholder/services/asset_service.go
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Phase 3: Trustee-Specific Endpoints

Goal: Dashboard, due diligence, fund management, and distribution authorization for Trustees.

Tasks

3A — Dashboard Home

- 3.1 GET /api/v1/stakeholder/trustee/dashboard — aggregate stats
 - Assets under trust count & total value
 - Pending milestone verifications
 - Pending fund release approvals
 - Pending distribution authorizations
 - Recent activity feed

3B — Due Diligence & Verification

- 3.2 DB migration: due_diligence_checklists table (asset_id, category, item, status, notes, verified_by, verified_at)
- 3.3 GET /api/v1/stakeholder/trustee/due-diligence/:asset_id — get checklist for asset
- 3.4 PUT /api/v1/stakeholder/trustee/due-diligence/:asset_id/item/:item_id — mark item complete, add notes
- 3.5 POST /api/v1/stakeholder/trustee/due-diligence/:asset_id/approve — approve asset for custody
- 3.6 POST /api/v1/stakeholder/trustee/due-diligence/:asset_id/reject — reject with reasons

3C — Fund Management & Release Approvals

3.7 DB migration: fund_release_requests table (id, asset_id, requester_org_id, amount, purpose, status, supporting_docs, reviewed_by, reviewed_at, notes)

3.8 GET /api/v1/stakeholder/trustee/fund-releases — list pending/all fund release requests

3.9 GET /api/v1/stakeholder/trustee/fund-releases/:request_id — request detail with docs

3.10 POST /api/v1/stakeholder/trustee/fund-releases/:request_id/approve — approve release

3.11 POST /api/v1/stakeholder/trustee/fund-releases/:request_id/reject — reject with reason

3D — Income Distribution Management

3.12 DB migration: distributions table (id, asset_id, proposed_by_org_id, amount, source, scheduled_date, status, authorized_by, authorized_at)

3.13 GET /api/v1/stakeholder/trustee/distributions — list distribution proposals

3.14 POST /api/v1/stakeholder/trustee/distributions/:dist_id/authorize — authorize distribution

3.15 POST /api/v1/stakeholder/trustee/distributions/:dist_id/reject — reject with reason

3.16 GET /api/v1/stakeholder/trustee/distributions/history — past distributions

Phase 4: Asset Custodian-Specific Endpoints

Goal: Account management, fund release execution, and compliance for Custodians.

Tasks

4A — Dashboard Home

4.1 GET /api/v1/stakeholder/custodian/dashboard — aggregate stats

- Segregated accounts count & balances
- Pending fund release requests
- Reconciliation status
- Compliance status

4B — Segregated Accounts

4.2 DB migration: segregated_accounts table (id, asset_id, custodian_org_id, account_type, account_name, balance, status, bank_details, created_at)

4.3 GET /api/v1/stakeholder/custodian/accounts — list accounts with balances

4.4 GET /api/v1/stakeholder/custodian/accounts/:account_id — account detail with transaction history

4C — Fund Release Processing

4.5 GET /api/v1/stakeholder/custodian/fund-releases — list Trustee-authorized releases pending execution

4.6 POST /api/v1/stakeholder/custodian/fund-releases/:request_id/execute — execute the release

4.7 PUT /api/v1/stakeholder/custodian/fund-releases/:request_id/status — update release status (processing, completed, failed)

4D — Compliance

4.8 DB migration: compliance_items table (id, org_id, category, requirement, status, due_date, completed_at)

4.9 GET /api/v1/stakeholder/custodian/compliance — compliance checklist

4.10 PUT /api/v1/stakeholder/custodian/compliance/:item_id — update compliance item status

Phase 5: Asset Manager-Specific Endpoints

Goal: Asset operations, revenue management, valuations, and reporting for Asset Managers.

Tasks

5A — Dashboard Home

5.1 GET /api/v1/stakeholder/asset-manager/dashboard — aggregate stats

- Assets managed count & total value
- Revenue collection status (current period, YTD)
- Pending milestone completions
- Pending Trustee approvals

5B — Assets Under Management

5.2 GET /api/v1/stakeholder/asset-manager/assets — list managed assets with operational details

5.3 PUT /api/v1/stakeholder/asset-manager/assets/:asset_id — update asset operational info

5C — Revenue Management

5.4 DB migration: revenue_records table (id, asset_id, manager_org_id, period, source, amount, status, collected_at)

5.5 GET /api/v1/stakeholder/asset-manager/revenue — revenue dashboard (current, YTD, by source)

5.6 POST /api/v1/stakeholder/asset-manager/revenue — record revenue collection

5.7 POST /api/v1/stakeholder/asset-manager/revenue/submit-distribution — submit revenue for distribution (creates distribution proposal for Trustee)

5D — Valuations

5.8 DB migration: asset_valuations table (id, asset_id, manager_org_id, valuation, methodology, valuation_date, report_url, status)

5.9 GET /api/v1/stakeholder/asset-manager/valuations/:asset_id — valuation history

5.10 POST /api/v1/stakeholder/asset-manager/valuations — submit new valuation

5.11 POST /api/v1/stakeholder/asset-manager/valuations/:id/request-independent — request independent valuation

5E — Fund Release Requests (from AM side)

5.12 POST /api/v1/stakeholder/asset-manager/fund-releases — create fund release request (goes to Trustee for approval)

5.13 GET /api/v1/stakeholder/asset-manager/fund-releases — view own requests and their status

5F — Reporting

5.14 POST /api/v1/stakeholder/asset-manager/reports — generate report (financial, operational)

5.15 GET /api/v1/stakeholder/asset-manager/reports — list generated reports

5.16 POST /api/v1/stakeholder/asset-manager/reports/:report_id/submit — submit report to Trustee

Phase 6: Cross-Functional Features

Goal: Document management, audit trail, communications, and profile settings.

Tasks

6A — Document Repository

6.1 DB migration: documents table (id, asset_id, uploaded_by_org_id, category, title, file_url, version, access_roles, created_at)

6.2 POST /api/v1/stakeholder/shared/documents — upload document

6.3 GET /api/v1/stakeholder/shared/documents — list documents (filtered by role access)

6.4 GET /api/v1/stakeholder/shared/documents/:doc_id — download/view document

6B — Activity & Audit Trail

6.5 DB migration: audit_logs table (id, user_id, org_id, role, action, entity_type, entity_id, details, created_at)

- 6.6 Create audit logging helper — call from all state-changing handlers
- 6.7 GET /api/v1/stakeholder/shared/audit-trail — list activities (filterable by date, role, action type, asset)

6C — Communications & Notifications

- 6.8 DB migration: notifications table (id, recipient_org_id, sender_org_id, type, title, message, related_entity_type, related_entity_id, read, created_at)
- 6.9 GET /api/v1/stakeholder/shared/notifications — list notifications
- 6.10 PUT /api/v1/stakeholder/shared/notifications/:id/read — mark as read
- 6.11 Auto-generate notifications on key actions (fund release approved, distribution authorized, milestone verified, etc.)

6D — User Profile

- 6.12 GET /api/v1/stakeholder/shared/profile — get current user profile with org and stakeholder info
 - 6.13 PUT /api/v1/stakeholder/shared/profile — update profile info
 - 6.14 PUT /api/v1/stakeholder/shared/profile/notifications — update notification preferences
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Phase 7: Technical Hardening

- 7.1 Run all DB migrations in order
 - 7.2 Add database indexes on frequently queried columns (asset_id, org_id, status, created_at)
 - 7.3 Enforce 2FA (via existing wallet linkage auth flow) on: fund release approvals, distribution authorizations, fund release execution
 - 7.4 Add Redis caching for dashboard aggregate queries
 - 7.5 Ensure all list endpoints support pagination (page, limit, sort, filters)
 - 7.6 Add Swagger annotations for all new endpoints
 - 7.7 Add request validation on all POST/PUT endpoints
 - 7.8 Security review: ensure no cross-role data leakage, validate org ownership on all entity access
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New Database Tables Summary

Table	Phase	Purpose
due_diligence_checklists	3	Trustee verification items per asset
fund_release_requests	3	Fund release request workflow

distributions	3	Income distribution proposals and authorizations
segregated_accounts	4	Custodian-managed bank accounts
compliance_items	4	Compliance tracking per org
revenue_records	5	Revenue collection records
asset_valuations	5	Valuation history per asset
documents	6	Document repository
audit_logs	6	Activity and audit trail
notifications	6	Inter-stakeholder notifications

Estimated Endpoint Count

Area	Endpoint s
Shared (assets, docs, audit, profile)	~10
Trustee	~12
Asset Custodian	~8
Asset Manager	~12
Total new endpoints	~42